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PREBOARD EXAMINATION - I (2025-26)

SUBJECT - PHYSICS

MM : 70 **CLASS-XII** **Time: 3:00 hours**

Section A - MCQ

Q 1. A uniform electric field pointing in positive X-direction exists in a region. Let A be the origin, B be the point on the X-axis at $x = +1$ cm and C be the point on the Y-axis at $y = +1$ cm. Then the potential at points A, B and C satisfy. [1]

A. $V_A < V_B$ B. $V_A > V_B$ C. $V_A < V_C$ D. $V_A > V_C$

Q 2. A galvanometer of resistance $6\,\Omega$ shows full scale deflection for a current of 0.2 A. The value of shunt to be used with this galvanometer to convert it into an ammeter of range $(0 - 5$ A$)$ is : [1]

A. $0.25\,\Omega$ B. $0.30\,\Omega$ C. $0.50\,\Omega$ D. $6.0\,\Omega$

Q 3. Calculate the mutual inductance between two coils if a current 10 A in the primary coil changes the flux by 500 Wb per turn in the secondary coil of 200 turns. [1]

A. 10 H B. 1×10^4 H C. 1×10^3 H D. 1×10^2 H

Q 4. A point object is placed at the centre of a glass sphere of radius 6 cm and refractive index 1.5. The distance of virtual image from the surface of the sphere is [1]

A. 2 cm B. 4 cm C. 6 cm D. 12 cm

Q 5. A battery supplies 0.9 A current through a $2\,\Omega$ resistor and 0.3 A current through a $7\,\Omega$ resistor when connected on by one. The internal resistance of the battery is : [1]

A. $2\,\Omega$ B. $1.2\,\Omega$ C. $1\,\Omega$ D. $0.5\,\Omega$

Q 6. The ratio of the energies of the hydrogen atom in its first to second excited state is: [1]

A. $\frac{1}{4}$ B. $\frac{4}{9}$ C. $\frac{9}{4}$ D. 4

Q 7. In a plane electromagnetic wave, the magnetic field oscillated sinusoidally with a frequency of 7.5×10^7 Hz, and amplitude 2.4×10^{-8} T. The amplitude of the oscillating electric field is : [1]

A. $1.8\,\text{Vm}^{-1}$ B. $3.2\,\text{Vm}^{-1}$ C. $7.2\,\text{Vm}^{-1}$ D. $8.0\,\text{Vm}^{-1}$

Q 8. Two point charges A and B , having charges $+Q$ and $-Q$ respectively , are placed at certain distance apart and force acting between them is F . If 25% charge of A is transferred to B , then the force between the charges becomes [1]

A. $\frac{9F}{16}$ B. $\frac{16F}{9}$ C. $\frac{4F}{3}$ D. F

Q 9. Two identical electrical dipoles, each consisting of charges $-q$ and $+q$ separated by distance d, are arranged in x-y plane such that their negative charges lie at the origin O and positive charges lie at points $(d,\ 0)$ and $(0,\ d)$ respectively. The net dipole moment of the system is : [1]

A. $-qd(\hat{i}+\hat{j})$ B. $qd(\hat{i}+\hat{j})$ C. $qd(\hat{i}-\hat{j})$ D. $qd(\hat{j}-\hat{i})$

Q 10. Lenz's law is essential for- [1]

A. conservation of energy
C. conservation of momentum
B. conservation of angular momentum
D. conservation of charge

Q 11. When two nuclei ($A \leq 10$) fuse together to form a heavier nucleus, the [1]

A. binding energy per nucleon increases.
B. binding energy per nucleon decreases.
C. binding energy per nucleon does not change.
D. total binding energy decrease.

Q 12. In a pure inductive circuit, the current- [1]

A. lags behind the applied emf by an angle π.
B. lags behind the applied emf by an angle $\frac{\pi}{2}$.
C. leads the applied emf by an angle $\frac{\pi}{2}$.
D. and applied emf are in same phase.

Q 13. Waves in decreasing order of their wavelength are [1]

A. radio waves, ultraviolet rays, visible rays, X-rays
B. radio waves, visible rays, infrared rays, X-rays
C. radio waves, infrared rays, visible rays, X-rays
D. X-rays, infrared rays, visible rays, radio waves

*Questions number **14 to 16** are **Assertion (A)** and **Reason (R)** type questions. Two statements are given - one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c) and (d) as given below.*

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(c) Assertion (A) is true, but Reason (R) is false.

(d) Assertion (A) is false and Reason (R) is also false.

Q 14. ***Assertion (A) :*** When three electric bulbs of power 200 W, 100 W and 50 W are connected in series to a source the power consumed by the 50 W bulb is maximum. [1]

Reason (R) : In series circuit, current is same through each bulb, but the potential difference across each bulb is different.

Q 15. ***Assertion (A) :*** The deflecting torque acting on a current carrying loop is zero when its plane is perpendicular to the direction of magnetic field. [1]

Reason (R) : The deflecting torque acting on a loop of magnetic moment $\vec{m}$ in a magnetic field $\vec{B}$ is given by the dot product of $\vec{m}$ and $\vec{B}$.

Q 16. ***Assertion (A) :*** There exists two angles of incidence for the same magnitude of deviation (except minimum deviation) by a prism kept in air. [1]

Reason (R) : In a prism kept in air, a ray is incident on first surface and emerges out of second surface. Now if another ray is incident on the second surface (of prism) along the previous emergent ray, then this ray emerges out of first surface along the previous incident ray. This is called principle o reversibility of light.

Section B - Very Short Answer Questions

Q 17. **(a)** Two small conducting balls A and B of radius r_1 and r_2 have charges q_1 and q_2 respectively. They are connected by a wire. Obtain the expression for charges on A and B, in equilibrium. [2]

OR

(b) Three point charges $1\,\mu C$, $-1\,\mu C$ and $2\,\mu C$ are kept at the vertices A, B and C respectively of an equilateral triangle of side 1 m. A_1, B_1, and C_1 are the midpoints of the sides AB, BC and CA respectively. Calculate the net amount of work done in displacing the charge from A to A_1, from B to B_1 and from C to C_1. [2]

Q 18. **(a)** Three capacitors each of capacitance 9 pF are connected in series. [2]

(i) What is the total capacitance of the combination?

(ii) What is the potential difference across each capacitor if the combination is connected to a 120 V supply?

OR

(b) Two point charges q_1 and q_2 in air are located at $\vec{r}_1$ and $\vec{r}_2$ respectively in a region of uniform external field $\vec{E}$. Obtain an expression for the electrostatic potential energy of the system. [2]

Q 19. **(a)** Write three points of differences between para-, dia- and ferro- magnetic materials, giving one example of each. [2]

OR

(b) **(i)** Define the term magnetic susceptibility and write its relation in terms of relative magnetic permeability. [2]

(ii) State Gauss's law in magnetism. How is it different from Gauss's law in electrostatics and why?

Q 20. **(a)** What is meant by the term 'displacement current'? Briefly explain how this current how this current is different form a conduction current. [2]

OR

(b) **(i)** The electric field of an electromagnetic wave passing through vacuum is represented as $E_x = E_0 \sin(kz - \omega t)$. Identify the parameter which is related to the (i) wavelength, and (ii) the frequency of the wave in above equation. [2]

(ii) Write two properties of a medium that determines the velocity of light in that medium.

Q 21. **(a)** The radius of ${}^{27}_{13}X$ nucleus is R. Then Find the radius of ${}^{125}_{53}Y$ nucleus. [2]

OR

(b) Draw a graph showing the variation of potential energy of pair of nucleons as a function of their separation. Indicate the region in which the nuclear force is (a) attractive and (b) repulsive. [2]

Section C - Short Answer Questions

Q 22. **(a)** **(i)** An electric dipole is held in a uniform electric field. Using suitable diagram show that it does not undergo any translatory motion. Derive the expression for the torque acting on it. [3]

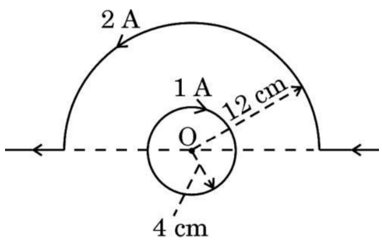
(ii) What would happen if the field in non-uniform ?

OR

(b) **(i)** Use Gauss's law to prove that the electric field at a point due to a uniformly charged infinite plane sheet is independent of the distance from it. [3]

(ii) Two parallel large thin metal sheets have equal surface densities 26.4×10^{-12} Cm^{-2} of opposite signs. Find the electric field between these sheets.

Q 23. **(a)** A current carrying circular loop and a straight wire bent partly in the form of a semicircle are placed as shown in the figure. Find the magnitude and direction of net magnetic field at point O. [3]



OR

(b) A wire AB is carrying a steady current of 12 A and is lying on the table. Another wire CD carrying 5 A is held directly above AB at a height of 1 mm. Find the mass per unit length of the wire CD so that it remains suspended at its position when left free. Give the direction of the current flowing in CD with respect to that in AB. [Take the value of $g = 10$ ms^{-2}] [3]

Q 24. **(a)** Draw the circuit diagram of a full wave rectifier using P – N junction diode. Explain its working and show the output and input waveforms. [3]

OR

(b) **(i)** Why is an intrinsic semiconductor deliberately converted into an extrinsic semiconductor by adding impurity atoms? [3]

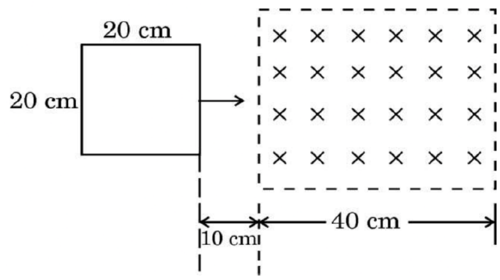
(ii) Explain briefly the two processes that occur in p-n junction region to create a potential barrier.

Q 25. **(a)** **(i)** Define self-inductance of a coil. Show that magnetic energy required to build up the current I in a coil of self-inductance L is given by $\frac{1}{2}LI^2$. [3]

(ii) The magnetic flux linked with a coil (in Wb) is given by the equation: $\phi = 5t^2 + 3t + 16$. Find The induced e.m.f. in the coil in the fourth second.

OR

(b) A square loop of side 20 cm starts moving at $t = 0$ with a velocity of 5 cm/s toward a region of uniform magnetic field as shown in the figure. Specify the time interval(s) during which induced emf is produced in the loop. [3]



Q 26. **(a)** Draw a neat ray diagram of refraction of light through equilateral prism and find the expression for refractive index of material of prism. Draw the graph showing the variation of angle of deviation with angle of incidence. [3]

OR

(b) **(i)** A concave lens made of material of refractive index (n_2) is held in a reference medium of refractive index (n_1). Trace the path of parallel beam of light passing through the lens when: [3]
(a) $n_1 = n_2$ (b) $n_1 < n_2$ (c) $n_1 > n_2$

(ii) Find the focal length of a combination of a convex lens of focal length 30 cm and a concave lens of focal length 20 cm in contact?

Q 27. **(a)** **(i)** Draw a plot of binding energy per nucleon (BE/A)versus mass number(A) for a large number of nuclei lying between. Using this graph , explain clearly how the energy is released in both the process of nuclear fission and nuclear fusion. [3]

(ii) Prove that nuclear density is constant for all nuclei.

OR

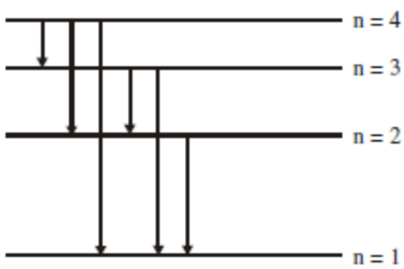
(b) **(i)** Define mass defect and binding energy? [3]

(ii) Calculate the energy required to dissociate a deuteron into its constituent particles (a proton and a neutron). Given: mass of deuteron = 2.014102 u, mass of proton = 1.007825 u, mass of neutron = 1.008665 u.

Q 28. **(a)** The figure shows energy level diagram of hydrogen atom. [3]

(i) Find out the transition which results in the emission of a photon of wavelength 496 nm

(ii) Which transition corresponds to the emission of radiation of maximum wavelength ? Justify your answer.



OR

(b) **(i)** Differentiate between 'distance of closest approach' and 'impact parameter'. [3]

(ii) Determine the distance of closest approach when an alpha particle o kinetic energy 3.95 MeV approaches a nucleus of $Z = 79$, stops and reverses its direction.

Section D - Long Answer Questions

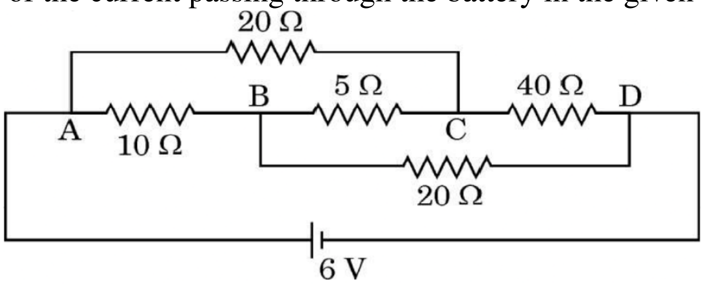
Q 29. **(a)** **(i)** On the basis of electron drift, derive an expression for resistivity of a conductor in terms of number density of free electrons and relaxation time. On what factors does resistivity of a conductor depend? [5]

(ii) Two conducting wires X and Y of same diameter but different materials are joined in series across a battery. If the number density of electrons in X is twice that in Y, find the ratio of drift velocity of electrons in the two wires.

OR

(b) **(i)** State Kirchhoff's rules. Draw a circuit diagram showing balancing of Wheatstone bridge. Use Kirchhoff's rules to obtain the balance condition in terms of the resistances of four arms of Wheatstone Bridge. [5]

(ii) Calculate the value of the current passing through the battery in the given circuit diagram.



Q 30. **(a)** **(i)** With a labelled diagram of an A.C generator briefly explain the working of the generator and derive the expressions for instantaneous emf. [5]

(ii) A circular coil of radius 8.0 cm and 20 turns is rotated about its vertical diameter with an angular speed of $50\,\text{rad}\,\text{s}^{-1}$ in a uniform horizontal magnetic field of magnitude 3×10^{-2} T. Obtain the **maximum and average emf induced** in the coil. If the coil forms a closed loop of resistance $10\,\Omega$, **calculate** I_{max} and I_{rms}.

OR

(b) **(i)** With a labelled diagram of a transformer briefly explain the working of the transformer and derive an expression for transformer ratio. [5]

(ii) An ac voltage $V_i = 140\sin(100\pi t)$ V is applied to the primary coil having 200 turns, of an ideal transformer and it supplies a power of 5 kW. If the secondary coil has 1000 turns, find :

(I) the output voltage,

(II) the instantaneous voltage across the secondary coil, and

(III) the current in the secondary coil. (Take $\sqrt{2} = 1.4$)

Q 31. **(a)** **(i)** Draw a labelled ray diagram to obtain the real image formed by an astronomical telescope in normal adjustment position. Define its magnifying power. [5]

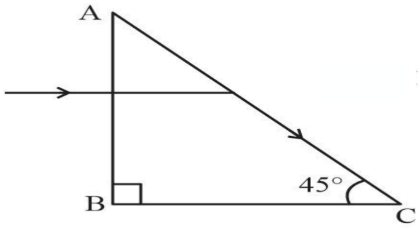
(ii) You are given three lenses of power 0.5 D, 4 D and 10 D to design a telescope.

(I) Which lenses should be used as objective and eyepiece? Justify your answer.

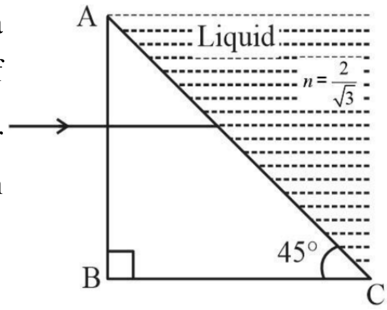
(II) Why is the aperture of the objective preferred to be large?

OR

(b) **(i)** A light ray entering a right-angled prism undergoes refraction at the face AC as shown in Fig. What is the refractive index of the material of the prism in fig. [5]



(ii) **(I)** If the side AC of the above prism is now surrounded by a liquid of refractive index $\frac{2}{\sqrt{3}}$ shown in Fig., determine if the light ray continues to graze along the interface AC or undergoes total internal reflection or undergoes refraction into the liquid.



(II) Draw the ray diagram to represent the path followed by the incident ray with the corresponding angle values. $\left(\text{Given: } \sin^{-1}\left(\frac{\sqrt{2}}{\sqrt{3}}\right) = 54.6^\circ\right)$

Section E - Case Study

Q 32. A pure semiconductor like Ge or Si, when doped with a small amount of suitable impurity, becomes an extrinsic semiconductor. In thermal equilibrium, the electron and hole concentration in it are related to the concentration of intrinsic charge carriers. A p-type or n-type semiconductor can be converted into a p-n junction by doping it with suitable impurity. Two processes, diffusion and drift take place during formation of a p-n junction. A semiconductor diode is basically a p-n junction with metallic contacts provided at the ends for the application of an external voltage. A p-n junction diode allows currents to pass only in one direction when it is forward biased. Due to this property, a diode is widely used to rectify alternating voltages, in half-wave or full wave configuration. [4]

(i) At a certain temperature in an intrinsic semiconductor, the electrons and holes concentration is $1.5 \times 10^{16}\ \mathrm{m^{-3}}$. When it is doped with a trivalent dopant, hole concentration increases to $4.5 \times 10^{22}\ \mathrm{m^{-3}}$. In the doped semiconductor, the concentration of electrons (n_e) will be :

A. $3 \times 10^{6}\ \mathrm{m^{-3}}$
B. $5 \times 10^{7}\ \mathrm{m^{-3}}$
C. $5 \times 10^{9}\ \mathrm{m^{-3}}$
D. $6.75 \times 10^{38}\ \mathrm{m^{-3}}$

(ii) Which of the following is a donor impurity atom for Ge?

A. Boron
B. Antimony
C. Aluminium
D. Indium

(iii) The doping of a semiconductor with donor impurity results in :

A. formation of a p-type semiconductor
B. formation of a new energy level just above the filled valance band
C. decrease in the concentration of electrons
D. formation of a new energy level just below the conduction band

(iv) An ac voltage $V = 0.5\sin(100\pi t)$ volt is applied, in turn, across a half-wave rectifier and a full-wave rectifier. The frequency of the output voltage across them respectively will be

A. 25 Hz, 50 Hz
B. 100 Hz, 100 Hz
C. 50 Hz, 100 Hz
D. 50 Hz, 25 Hz

Q 33. A lens is a transparent optical medium bounded by two surfaces; at least one of which should be spherical. Applying the formula of image formation by a single spherical surface successively at the two surfaces of a thin lens, a formula known as lens maker's formula and hence the basic lens formula can be obtained. The focal length (or power) of a lens depends on the radii of its surfaces and the refractive index of its material with respect to the surrounding medium. The refractive index of a material depends on the wavelength of light used. Combination of lenses helps us to obtain diverging or converging lenses of desired power and magnification. [4]

(i) A thin converging lens of focal length 20 cm and a thin diverging lens of focal length 15 cm are placed coaxially in contact. The power of the combination is

A. $\frac{-5}{6}$ D
B. $\frac{4}{3}$ D
C. $\frac{-5}{3}$ D
D. $\frac{3}{2}$ D

(ii) The radii of curvature of two surfaces of a convex lens are R and $2R$. If the focal length of this lens is $\frac{4}{3}R$ the refractive index of the material of the lens is :

A. $\frac{5}{3}$
B. $\frac{4}{3}$
C. $\frac{3}{2}$
D. $\frac{7}{5}$

(iii) A beam of light coming parallel to the principal axis of a convex lens L_1 of focal length 16 cm is incident on it. Another convex lens L_2 of focal length 12 cm is placed coaxially at a distance 40 cm from L_1. The nature and distance of the final image from L_2 will be

A. real, 24 cm
B. virtual, 12 cm
C. real, 32 cm
D. virtual, 18 cm

(iv) A point object is kept 60 cm in front of a spherical convex surface ($n = 1.5$, radius of curvature 40 cm). The image formed is

A. real, at a distance 1.8 m from the surface.
B. virtual, at a distance 1.8 m from the surface.
C. real, at a distance 3.6 m from the surface.
D. virtual, at a distance 3.6 m from the surface.

End of exam.